Oliver H. E. Philcox, Ph.D.

Physics & Astrophysics Building, 452 Lomita Mall, Stanford, CA 94305, USA

Website: oliverphilcox.github.io

Email: ohep2@cantab.ac.uk, ophilcox@stanford.edu

CURRENT POSITION

Department of Physics, Stanford University, Stanford, USA *2025 - Present*
Assistant Professor

Leinweber Institute for Theoretical Physics at Stanford, *Faculty*
Kavli Institute for Particle Astrophysics and Cosmology, *Senior Member*

PREVIOUS POSITIONS

Simons Society of Fellows, New York, USA *2022 - 2025*
Junior Fellow

Department of Physics, Columbia University, New York, USA *2022 - 2025*
Postdoctoral Research Scientist

Mentors: Prof. J. Colin Hill & Prof. Lam Hui

EDUCATION

Department of Astrophysical Sciences, Princeton University, USA *2019 - 2022*
Ph.D. in Astrophysics (2022)

Thesis: 'Probing Fundamental Cosmology with Galaxy Surveys'

Thesis Advisors: David N. Spergel & Matias Zaldarriaga

M.A. in Astrophysics (2020)

Center for Astrophysics | Harvard & Smithsonian, Cambridge, USA *2018 - 2019*
Pre-Doctoral Student, *Herchel-Smith Scholar*

Advisor: Daniel J. Eisenstein

Institute of Astronomy, University of Cambridge, UK *2017 - 2018*
M.Sci. in Astrophysics

Part III: 1st Class (Rank 1/28, 97%)

Thesis: 'Detection and Removal of B-mode CMB Dust Foregrounds with Signatures of Statistical Anisotropy'

Thesis Advisor: Blake D. Sherwin

Emmanuel College, University of Cambridge, UK *2014 - 2017*
M.A. (Cantab.) in Natural Sciences, *Senior Scholar*

Parts IA, IB, II: 1st Class (Rank 1/20, 90%)

LONG-TERM ACADEMIC VISITS

Center for Computational Astrophysics Jul. 2021 - Aug. 2025
Guest Researcher *New York, USA*

Institute for Advanced Study Sep. 2020 - Jul. 2022
Visiting Graduate Student with Matias Zaldarriaga *Princeton, USA*

Max-Planck Institute for Astrophysics Aug. - Sep. 2020
Visiting Graduate Student with Eiichiro Komatsu *Munich, Germany*

Department of Applied Mathematics and Theoretical Physics May - Jul. 2020
Visiting Graduate Student with Blake D. Sherwin *Cambridge, UK*

Max-Planck-Institut für Astronomie Jul. - Sep. 2017
Summer Intern with Jan Rybizki *Heidelberg, Germany*

Center for Astrophysics | Harvard & Smithsonian Jun. - Aug. 2016
Undergraduate Research Fellow with Ákos Bogdán *Cambridge, USA*

PUBLICATION LIST

** = Author list alphabetized*

Major Author

1. Smith, A. F., Ivanov, M. M., de Belsunce, R., Chen, S.-F., Chudaykin, A., **Philcox, O. H. E.**, "$w_0 w_a$ or bc: DESI constraints on relative baryon-CDM perturbations", (arXiv).
2. *Chudaykin, A., Ivanov, M. M., Kallosh, R., Linde, A., **Philcox, O. H. E.**, Yamada, Y., "Inflation, Open Universes, and Dark Energy", (arXiv).
3. Jacobus, C., Tucci, B., **Philcox, O. H. E.**, "Emulating Cosmic Structure Formation with a Lagrangian Neural Cellular Automaton", (arXiv).
4. **Philcox, O. H. E.**, "Dissecting the Scalar Cosmological Collider with the Cosmic Microwave Background", (arXiv).
5. Yang, S.-X., **Philcox, O. H. E.**, "Searching for Folded Primordial Non-Gaussianity with Galaxy Surveys", (arXiv).
6. **Philcox, O. H. E.**, "The Limitations of Cosmological Collider Analyses", (arXiv).
7. Jense, H. T., El-Haj, L. H. A., Hill, J. C., **Philcox, O. H. E.**, "Probing inflationary particle production with the CMB power spectrum", (arXiv).
8. *Gallardo, P., Pardo, K., **Philcox, O. H. E.**, Battaglia, N., Battistelli, E., Bean, R., *et al.*, "Test of the Gravitational Force Law on Cosmological Scales Using the Kinematic Sunyaev-Zeldovich Effect", *Phys. Rev. Lett.* **136**, 151002 (2026) (arXiv).
9. *Chen, S.-F., DeRose, J., Ivanov, M. M., **Philcox, O. H. E.**, "Cosmic Shear in Effective Field Theory at Two-Loop Order: Revisiting S_8 in Dark Energy Survey Data", (arXiv).
10. **Philcox, O. H. E.**, "What Shape is the Inflationary Bispectrum?", (arXiv).
11. ***Philcox, O. H. E.**, Pimentel, G. L., Yang, C., "Searching for unparticles with the cosmic microwave background", *Phys. Rev. D* **113**, 123529 (2026) (arXiv).
12. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1. V. Cosmological constraints with simulation-based priors", *Phys. Rev. D* **114**, 063527 (2026) (arXiv).
13. Ivanov, M. M., Sullivan, J. M., de Belsunce, R., Chen, S.-F., Chudaykin, A., Maus, M., *et al.* (inc. **Philcox, O. H. E.**), "Reanalyzing DESI DR1: IV. Percent-level cosmological constraints from combined probes and robust evidence for the normal neutrino mass hierarchy", *Phys. Rev. D* **114**, 063526 (2026) (arXiv).
14. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1. III. Constraints on inflation from galaxy power spectra and bispectra", *Phys. Rev. D* **113**, 063552 (2026) (arXiv).
15. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1. II. Constraints on dark energy, spatial curvature, and neutrino masses", *Phys. Rev. D* **113**, 123506 (2026) (arXiv).
16. **Philcox, O. H. E.**, Zhong, K., Sirletti, S. S., "Separating the Inseparable: Constraining Arbitrary Primordial Bispectra with Cosmic Microwave Background Data", *submitted to Phys. Rev. D* (arXiv).
17. El-Haj, L. H. A., **Philcox, O. H. E.**, Hill, J. C., "Constraining inflationary particle production with CMB polarization", *Phys. Rev. D* **113**, 103518 (2026) (arXiv).

18. *Bakx, T., Ivanov, M. M., **Philcox, O. H. E.**, Vlah, Z., "One-loop galaxy bispectrum: Consistent theory, efficient analysis with COBRA, and implications for cosmological parameters", *Phys. Rev. D* **114**, 063532 (2026) (arXiv).

19. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, "Reanalyzing DESI DR1. I. ΛCDM constraints from the power spectrum and bispectrum", *Phys. Rev. D* **113**, 063502 (2026) (arXiv).

20. ***Philcox, O. H. E.**, Silverstein, E., Torroba, G., "Quantum stress-energy at timelike boundaries: Testing a new beyond-ΛCDM parameter with cosmological data", *Phys. Rev. D* **113**, 043548 (2026) (arXiv).

21. **Philcox, O. H. E.**, Hill, J. C., "Integrated Sachs-Wolfe lensing bispectrum and trispectrum", *Phys. Rev. D* **111**, 123554 (2025) (arXiv).

22. Goldstein, S., **Philcox, O. H. E.**, Fondi, E., Coulton, W. R., "Wonderings on wiggly bispectra: Nonlinear evolution and reconstruction of oscillations in the squeezed bispectrum", *Phys. Rev. D* **112**, 083503 (2025) (arXiv).

23. Zhang, G., Modi, C., **Philcox, O. H. E.**, "Modeling galaxy surveys with hybrid simulation-based inference", *Phys. Rev. D* **113**, 043518 (2026) (arXiv).

24. **Philcox, O. H. E.**, "Searching for inflationary physics with the CMB trispectrum. III. Constraints from *Planck*", *Phys. Rev. D* **111**, 123534 (2025) (arXiv).

25. **Philcox, O. H. E.**, "Searching for inflationary physics with the CMB trispectrum. II. Code and validation", *Phys. Rev. D* **111**, 123533 (2025) (arXiv).

26. **Philcox, O. H. E.**, "Searching for inflationary physics with the CMB trispectrum. I. Primordial theory and optimal estimators", *Phys. Rev. D* **111**, 123532 (2025) (arXiv).

27. ***Philcox, O. H. E.**, Shiraishi, M., "Non-Gaussianity beyond the scalar sector: A search for tensor and mixed tensor-scalar bispectra with Planck data", *Phys. Rev. D* **111**, 123502 (2025) (arXiv).

28. Goldstein, S., **Philcox, O. H. E.**, Hill, J. C., Hui, L., "Intermediate mass-range particles from small scales: Nonperturbative techniques for cosmological collider physics from large-scale structure surveys", *Phys. Rev. D* **110**, 083516 (2024) (arXiv).

29. *Coulton, W. R., **Philcox, O. H. E.**, Villaescusa-Navarro, F., "The Impact of Non-Gaussian Primordial Tails on Cosmological Observables", *submitted to Phys. Rev. D* (arXiv).

30. *Chen, S.-F., Ivanov, M. M., **Philcox, O. H. E.**, Wenzl, L., "Suppression without Thawing: Constraining Structure Formation and Dark Energy with Galaxy Clustering", *Phys. Rev. Lett.* **133**, 231001 (2024) (arXiv).

31. **Philcox, O. H. E.**, Kumar, S., Hill, J. C., "Searching for inflationary particle production in Planck data", *Phys. Rev. D* **111**, 103523 (2025) (arXiv).

32. **Philcox, O. H. E.**, Flöss, T., "Suite of optimal and efficient power spectrum and bispectrum estimators for large-scale structure analyses", *Phys. Rev. D* **112**, 063507 (2025) (arXiv).

33. Cabass, G., **Philcox, O. H. E.**, Ivanov, M. M., Akitsu, K., Chen, S.-F., Simonović, M., *et al.*, "BOSS constraints on massive particles during inflation: The cosmological collider in action", *Phys. Rev. D* **111**, 063510 (2025) (arXiv).

34. de Belsunce, R., **Philcox, O. H. E.**, Irsic, V., McDonald, P., Guy, J., Palanque-Delabrouille, N., "The 3D Lyman-α forest power spectrum from eBOSS DR16", *MNRAS* **533**, 3756 – 3770 (2024) (arXiv).

35. **Philcox, O. H. E.**, Ereza, J., "Could Sample Variance be Responsible for the Parity-Violating Signal Seen in the BOSS Galaxy Survey?", *Phil. Trans. Roy. Soc. A* **383**, 2290 (2025) (arXiv).

36. ***Philcox, O. H. E.**, Shiraishi, M., "Testing graviton parity and Gaussianity with Planck T-, E-, and B-mode bispectra", *Phys. Rev. D* **109**, 063522 (2024) (arXiv).

37. Goldstein, S., **Philcox, O. H. E.**, Hill, J. C., Esposito, A., Hui, L., "Consistently constraining $f_{\rm NL}$ with the squeezed lensing bispectrum using consistency relations", *Phys. Rev. D* **109**, 043515 (2024) (arXiv).

38. *Modi, C., **Philcox, O. H. E.**, "Hybrid SBI or How I Learned to Stop Worrying and Learn the Likelihood", *submitted to Phys. Rev. Lett.* (arXiv).

39. **Philcox, O. H. E.**, König, M. J., Alexander, S., Spergel, D. N., "What can galaxy shapes tell us about physics beyond the standard model?", *Phys. Rev. D* **109**, 063541 (2024) (arXiv).

40. ***Philcox, O. H. E.**, Shiraishi, M., "Testing parity symmetry with the polarized cosmic microwave background", *Phys. Rev. D* **109**, 083514 (2024) (arXiv).

41. *Coulton, W. R., **Philcox, O. H. E.**, Villaescusa-Navarro, F., "Signatures of a parity-violating universe", *Phys. Rev. D* **109**, 023531 (2024) (arXiv).

42. **Philcox, O. H. E.**, "Optimal estimation of the binned mask-free power spectrum, bispectrum, and trispectrum on the full sky: Tensor edition", *Phys. Rev. D* **108**, 063506 (2023) (arXiv).

43. *Ivanov, M. M., **Philcox, O. H. E.**, "Measuring H_0 with Spectroscopic Surveys", *chapter in "Hubble Constant Tension" (Eds. Di Valentino, E. and Brout, D., Springer, Singapore)* (arXiv).

44. **Philcox, O. H. E.**, "Do the CMB Temperature Fluctuations Conserve Parity?", *Phys. Rev. Lett.* **131**, 181001 (2023) (arXiv).

45. **Philcox, O. H. E.**, "Optimal estimation of the binned mask-free power spectrum, bispectrum, and trispectrum on the full sky: Scalar edition", *Phys. Rev. D* **107**, 123516 (2023) (arXiv).

46. Creque-Sarbinowski, C., Alexander, S., Kamionkowski, M., **Philcox, O. H. E.**, "Parity-violating trispectrum from Chern-Simons gravity", *JCAP* 11 029 (2023) (arXiv).

47. Surrao, K. M., **Philcox, O. H. E.**, Hill, J. C., "Accurate estimation of angular power spectra for maps with correlated masks", *Phys. Rev. D* **107**, 083521 (2023) (arXiv).

48. Ivanov, M. M., **Philcox, O. H. E.**, Cabass, G., Nishimichi, T., Simonović, M., Zaldarriaga, M., "Cosmology with the galaxy bispectrum multipoles: Optimal estimation and application to BOSS data", *Phys. Rev. D* **107**, 083515 (2023) (arXiv).

49. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonovic, M., Zaldarriaga, M., "Constraining single-field inflation with MegaMapper", *Phys. Lett. B* **841**, 137912 (2023) (arXiv).

50. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, "Colliders and ghosts: Constraining inflation with the parity-odd galaxy four-point function", *Phys. Rev. D* **107**, 023523 (2023) (arXiv).

51. Goldstein, S., Esposito, A., **Philcox, O. H. E.**, Hui, L., Hill, J. C., Scoccimarro, R., *et al.*, "Squeezing $f_{\rm NL}$ out of the matter bispectrum with consistency relations", *Phys. Rev. D* **106**, 123525 (2022) (arXiv).

52. ***Philcox, O. H. E.**, Torquato, S., "Disordered Heterogeneous Universe: Galaxy Distribution and Clustering across Length Scales", *Phys. Rev. X* **13**, 011038 (2023) (arXiv).

53. **Philcox, O. H. E.**, Johnson, M. C., "Novel cosmological tests from combining galaxy lensing and the polarized Sunyaev-Zel'dovich effect", *Phys. Rev. D* **106**, 083501 (2022) (arXiv).

54. **Philcox, O. H. E.**, "Probing parity violation with the four-point correlation function of BOSS galaxies", *Phys. Rev. D* **106**, 063501 (2022) (arXiv).

55. **Philcox, O. H. E.**, Ivanov, M. M., Cabass, G., Simonović, M., Zaldarriaga, M., Nishimichi, T., "Cosmology with the redshift-space galaxy bispectrum monopole at one-loop order", *Phys. Rev. D* **106**, 043530 (2022) (arXiv).

56. **Philcox, O. H. E.**, Farren, G. S., Sherwin, B. D., Baxter, E. J., Brout, D. J., "Determining the Hubble constant without the sound horizon: A 3.6% constraint on H_0 from galaxy surveys, CMB lensing, and supernovae", *Phys. Rev. D* **106**, 063530 (2022) (arXiv).

57. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., "Constraints on multifield inflation from the BOSS galaxy survey", *Phys. Rev. D* **106**, 043506 (2022) (arXiv).

58. *Cabass, G., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., "Constraints on Single-Field Inflation from the BOSS Galaxy Survey", *Phys. Rev. Lett.* **129**, 021301 (2022) (arXiv).

59. *Farren, G. S., **Philcox, O. H. E.**, Sherwin, B. D., "Determining the Hubble constant without the sound horizon: Perspectives with future galaxy surveys", *Phys. Rev. D* **105**, 063503 (2022) (arXiv).

60. **Philcox, O. H. E.**, Ivanov, M. M., "BOSS DR12 full-shape cosmology: ΛCDM constraints from the large-scale galaxy power spectrum and bispectrum monopole", *Phys. Rev. D* **105**, 043517 (2022) (arXiv).

61. Ivanov, M. M., **Philcox, O. H. E.**, Nishimichi, T., Simonović, M., Takada, M., Zaldarriaga, M., "Precision analysis of the redshift-space galaxy bispectrum", *Phys. Rev. D* **105**, 063512 (2022) (arXiv).

62. Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., Zaldarriaga, M., Nischimichi, T., Takada, M., "Cosmological constraints without nonlinear redshift-space distortions", *Phys. Rev. D* **105**, 043531 (2022) (arXiv).

63. **Philcox, O. H. E.**, Hou, J., Slepian, Z., "A First Detection of the Connected 4-Point Correlation Function of Galaxies Using the BOSS CMASS Sample", *submitted to Phys. Rev. D* (arXiv).

64. **Philcox, O. H. E.**, "Cosmology without window functions. II. Cubic estimators for the galaxy bispectrum", *Phys. Rev. D* **104**, 123529 (2021) (arXiv).

65. ***Philcox, O. H. E.**, Slepian, Z., "Efficient computation of N-point correlation functions in D dimensions", *PNAS* **119**, 33 (2022) (arXiv).

66. **Philcox, O. H. E.**, Slepian, Z., Hou, J., Warner, C., Cahn, R. N., Eisenstein, D. J., "encore: an O (Ng2) estimator for galaxy N-point correlation functions", *MNRAS* **509**, 2457 – 2481 (2021) (arXiv).

67. ***Philcox, O. H. E.**, Slepian, Z., "An exact integral-to-sum relation for products of Bessel functions", *Proc. Roy. Soc. A* **477**, 2253 (2021) (arXiv).

68. Slepian, Z., **Philcox, O. H. E.**, "A uniform spherical goat (problem): explicit solution for homologous collapse's radial evolution in time", *MNRAS* **522**, L42-L45 (2023) (arXiv).

69. ***Philcox, O. H. E.**, Slepian, Z., "Beyond the Yamamoto approximation: Anisotropic power spectra and correlation functions with pairwise lines of sight", *Phys. Rev. D* **103**, 123509 (2021) (arXiv).

70. **Philcox, O. H. E.**, "Cosmology without window functions: Quadratic estimators for the galaxy power spectrum", *Phys. Rev. D* **103**, 103504 (2021) (arXiv).

71. **Philcox, O. H. E.**, Aviles, A., Massara, E., "Modeling the Marked Spectrum of Matter and Biased Tracers in Real- and Redshift-Space", *JCAP* 03 038 (2021) (arXiv).

72. **Philcox, O. H. E.**, Ivanov, M. M., Zaldarriaga, M., Simonovic, M., Schmittfull, M., "Fewer Mocks and Less Noise: Reducing the Dimensionality of Cosmological Observables with Subspace Projections", *Phys. Rev. D* **103**, 043508 (2021) (arXiv).

73. **Philcox, O. H. E.**, Sherwin, B. D., Farren, G. S., Baxter, E. J., "Determining the Hubble Constant without the Sound Horizon: Measurements from Galaxy Surveys", *Phys. Rev. D* **103**, 023538 (2021) (arXiv).

74. **Philcox, O. H. E.**, Massara, E., Spergel, D. N., "What does the marked power spectrum measure? Insights from perturbation theory", *Phys. Rev. D* **102**, 043516 (2020) (arXiv).

75. **Philcox, O. H. E.**, "A faster Fourier transform? Computing small-scale power spectra and bispectra for cosmological simulations in $\mathcal{O}(N^2)$ time", *MNRAS* **501**, 4004 – 4034 (2021) (arXiv).

76. ***Philcox, O. H. E.**, Spergel, D. N., Villaescusa-Navarro, F., "Effective halo model: Creating a physical and accurate model of the matter power spectrum and cluster counts", *Phys. Rev. D* **101**, 123520 (2020) (arXiv).

77. **Philcox, O. H. E.**, Ivanov, M. M., Simonović, M., Zaldarriaga, M., "Combining Full-Shape and BAO Analyses of Galaxy Power Spectra: A 1.6% CMB-independent constraint on H_0", *JCAP* 05 032 (2020) (arXiv).

78. **Philcox, O. H. E.**, Eisenstein, D. J., "Computing the Small-Scale Galaxy Power Spectrum and Bispectrum in Configuration-Space", *MNRAS* **492**, 1214 – 1242 (2020) (arXiv).

79. **Philcox, O. H. E.**, Eisenstein, D. J., "Estimating Covariance Matrices for Two- and Three-Point Correlation Function Moments in Arbitrary Survey Geometries", *MNRAS* **490**, 5931 – 5951 (2019) (arXiv).

80. ***Philcox, O. H. E.**, Rybizki, J., "Inferring Galactic Parameters from Chemical Abundances: A Multi-Star Approach", *ApJ* **887**, 9 (2019) (arXiv).

81. **Philcox, O. H. E.**, Eisenstein, D. J., O'Connell, R., Wiegand, A., "rascalc: a jackknife approach to estimating single- and multitracer galaxy covariance matrices", *MNRAS* **491**, 3290 – 3317 (2020) (arXiv).

82. ***Philcox, O. H. E.**, Sherwin, B. D., van Engelen, A., "Detection and Removal of B-mode Dust Foregrounds with Signatures of Statistical Anisotropy", *MNRAS* **479**, 5577 – 5595 (2018) (arXiv).

83. **Philcox, O. H. E.**, Rybizki, J., Gutcke, T., "On the Optimal Choice of Nucleosynthetic Yields, Initial Mass Function, and Number of SNe Ia for Chemical Evolution Modeling", *ApJ* **861**, 40 (2018) (arXiv).

Contributing Author

84. *Besuner, R., Dey, A., Drlica-Wagner, A., Ebina, H., Fernandez Moroni, G., Ferraro, S., *et al.* (inc. **Philcox, O. H. E.**), "The Spectroscopic Stage-5 Experiment", *White Paper* (arXiv).

85. *Krause, E., Kobayashi, Y., Salcedo, A. N., Ivanov, M. M., Abel, T., Akitsu, K., *et al.* (inc. **Philcox, O. H. E.**), "A Parameter-Masked Mock Data Challenge for Beyond-Two-Point Galaxy Clustering Statistics", *ApJ* **990**, 99 (2025) (arXiv).

86. Rogers, K. K., Hložek, R., Laguë, A., Ivanov, M. M., **Philcox, O. H. E.**, Cabass, G., *et al.*, "Ultra-light axions and the S_8 tension: joint constraints from the cosmic microwave background and galaxy clustering", *JCAP* 06 023 (2023) (arXiv).

87. *Abdalla, E., Franco Abellán, G., Aboubrahim, A., Agnello, A., Akarsu, O., Akrami, Y., *et al.* (inc. **Philcox, O. H. E.**), "Cosmology intertwined: A review of the particle physics, astrophysics, and cosmology associated with the cosmological tensions and anomalies", *JHEA* **34**, 49 – 211 (2022) (arXiv).

88. Villaescusa-Navarro, F., Genel, S., Anglés-Alcázar, D., Perez, L. A., Villanueva-Domingo, P., Wadekar, D., *et al.* (inc. **Philcox, O. H. E.**), "The CAMELS Project: Public Data Release", *ApJS* **265**, 54 (2023) (arXiv).

89. Hou, J., Cahn, R. N., **Philcox, O. H. E.**, Slepian, Z., "Analytic Gaussian covariance matrices for galaxy N-point correlation functions", *Phys. Rev. D* **106**, 043515 (2022) (arXiv).

90. Schmittfull, M., Simonović, M., Ivanov, M. M., **Philcox, O. H. E.**, Zaldarriaga, M., "Modeling Galaxies in Redshift Space at the Field Level", *JCAP* 05 059 (2021) (arXiv).

91. Villaescusa-Navarro, F., Anglés-Alcázar, D., Genel, S., Spergel, D. N., Somerville, R. S., Dave, R., *et al.* (inc. **Philcox, O. H. E.**), "The CAMELS project: Cosmology and Astrophysics with MachinE Learning Simulations", *ApJ* **915**, 71 (2021) (arXiv).

92. Wang, Y., Zhao, G.-B., Zhao, C., **Philcox, O. H. E.**, Alam, S., Tamone, A., *et al.*, "The clustering of the SDSS-IV extended Baryon Oscillation Spectroscopic Survey DR16 luminous red galaxy and emission line galaxy samples: cosmic distance and structure growth measurements using multiple tracers in configuration space", *MNRAS* **498**, 3470 – 3483 (2020) (arXiv).

93. *Chudaykin, A., Ivanov, M. M., **Philcox, O. H. E.**, Simonović, M., "Nonlinear perturbation theory extension of the Boltzmann code CLASS", *Phys. Rev. D* **102**, 063533 (2020) (arXiv).

Other Works

94. **Philcox, O. H. E.**, "An Introduction to the EFTofLSS", *Lecture Notes*.

SELECTED RESEARCH TALKS

* = *Virtual Talk*

2026 Cosmic Rhythms, Rio de Janeiro, *Conference*
New Frontiers in Cosmology, A Coruña, *Conference*
National Taiwan University, *Physics Colloquium*
*CosmoVerse Seminar, *Webinar* (Invited Talk)
SLAC National Laboratory, *Colloquium*
Max Planck-IAS-NTU Center Kick-Off, Princeton, *Conference* (Invited Talk)
Laboratoire d'Annecy de Physique Théorique, Annecy, France, *Seminar*
CERN, Geneva, Switzerland *Seminar*
New Synergies in Multiprobe Cosmology, KITP, Santa Barbara, *Conference*

2025 41st Institut d'Astrophysique de Paris Symposium, *Conference* (Invited Talk)
Galileo Galilei Institute for Theoretical Physics, *Symposium* (Invited Plenary)
21st Rencontres de Vietnam, *Conference* (Invited Talk)
Understanding Cosmological Observations, Benasque, *Conference* (Invited Talk)
GR24 & Amaldi16, *Conference* (Invited Plenary)
Columbia University, *Physics Colloquium*
Cosmological Probes of New Physics, Notre Dame, *Conference* (Invited Discussion)
Brown University, *Theoretical Physics Seminar*
Princeton University, *Cosmology Seminar*
Particles vs. New Probes, Flatiron Institute, *Conference* (Invited Plenary)
Boston University, *High-Energy Physics Seminar*

2024 21st Century Cosmology, Ashoka University, *Conference* (Invited Plenary)
Cosmology in the Adriatic, Split, *Conference*
University of Edinburgh, *Conference*
Lawrence Berkeley National Laboratory, *Workshop* (Invited Plenary)
University of Cambridge, *Astronomy Colloquium*
Royal Society, London, *Discussion Meeting* (Invited Talk)
Harvard University, *Cosmology Seminar*
Massachusetts Institute of Techology, *Cosmology Seminar*
58th Rencontres de Moriond, *Conference*
Stanford University, *Physics Colloquium*
Perimeter Institute for Theoretical Physics, *Colloquium*

2023 Taipei Institute of Astronomy & Astrophysics, *Conference* (Invited Talk)
Max-Planck-Institute for Nuclear Physics, Heidelberg, *Particle Physics Seminar*
ICTP Trieste, *Workshop*
University of Montreal, *Astrophysics Seminar*
*Parity-Violation from Home, *Conference*
Ohio State University, *Cosmology and Astro-Particle Physics Seminar*
*Early Universe / AliCPT Forum, University of Science and Technology of China, *Webinar*
Sexten Center for Astrophysics, *Workshop* (Invited Plenary)
Donostia International Physics Center, *Workshop* (Invited Talk)
University of Pennsylvania, *PDT Partners Retreat* (Invited Talk)
Yukawa Institute for Theoretical Physics, Kyoto, *Conference* (Invited Talk)
Kavli IPMU, Tokyo, *Astronomy Seminar*
Stony Brook, *Cosmology Seminar*
Cosmology on Safari, *Conference*
Johns Hopkins University, *Particle Physics Seminar*
University of Maryland, *Particle Physics Seminar*
*Copernicus Series, *Cosmology Webinar*
*University of Oxford, *Cosmology Seminar*

OUTREACH TALKS

2026 *Astronomy on Tap*, San Francisco (Public Talk)
Discover Our Universe, Kavli Institute for Particle Astrophysics & Cosmology (Public Lecture)

2025 *Stanford University Undergraduate Physics Club* (Faculty Lunch Talk)
Taste of Science, New York (Public Lecture)

2023 *Into The Unknown* (Public Podcast)

TEACHING & LECTURING

2026 Cosmology, *Stanford University* (Physics 361)
Croucher Advanced Study Institute on Primordial Non-Gaussianity, Hong Kong (Videos: Part I, Part II)

2024 Essential Cosmology for the Next Generation IX, Mexico, *Winter School* (Invited Lecture Series)

MEDIA

1. "Gravity follows Newton and Einstein's rules, even at cosmic scales", *Simons Foundation*, 15 Apr. 2026.
2. "Scientists detect mysterious suppression in cosmic structure growth", *Phys.org*, 3 Jan. 2025.
3. "Universe's 'Cosmological Collider' Lands 3 Scientists $100,000 Physics Prize", *Live Science*, 14 Sep. 2023.
4. "What Happened Right After the Universe Began?", *Simons Foundation*, 30 Aug. 2023.
5. "The Cosmos as a Colloid", *Physics Magazine*, 14 Mar. 2023.
6. "Pinpoint Simulations Provide Perspective on Universe Structure", *IAS News* & *Phys.org*, 14 Mar. 2023.
7. "Spatial Pattens In Distribution of Galaxies", *Princeton News* & *ScienceDaily*, 14 Mar. 2023.
8. "Is the Universe Asymmetrical?", *Columbia News*, 27 Feb. 2023.
9. "Do We Live in a Mirror Universe?", *Into The Unknown Podcast*, 26 Jan. 2023.
10. "Asymmetry Detected in the Distribution of Galaxies", *Quanta*, 5 Dec. 2022.
11. "The Universe is Surprisingly Lopsided and We Don't Know Why", *New Scientist*, 18 Jun. 2022.

CURRENT GRANTS

2026 NSF PHY-2609396 (PI; $140 000): *Probing High-Energy Physics with the Galaxy Bispectrum*

PRIZES

2024 New Horizons in Physics Prize, *Breakthrough Prize Foundation*

2023 Buchalter Cosmology Prize (First Prize)

2022 Simons Society of Fellows (Junior Fellowship)
NHFP Einstein Fellowship, *declined*
LBL Chamberlain Fellowship, *declined*
Cambridge Kavli Fellowship, *declined*

2018 Herchel-Smith Scholarship, *Cambridge* → *Harvard*
Institute of Astronomy Prize, *Cambridge*

2017 Holgate Pollard Memorial Prize, *Cambridge*

PROFESSIONAL ACTIVITIES

Referee	MNRAS (2020−), JCAP (2020−), MPLA (2021−), Phys. Rev. Lett. (2022−) Phys. Rev. D (2022−), ApJS (2022−), Phys. Dark Univ. (2023−)	
Conferences	Fundamental Physics from the Primordial Era with Galaxy Surveys (Aspen 2026, SOC) Large-Scale Parity Violation (Taiwan 2023, SOC)	
Review Panels	NASA Astrophysics Theory Program (2023)	
Mentor	*Dhayaa Anbajagane*	NASA Einstein Fellow (2026−)
	Beatriz Tucci	Stanford Postdoctoral Fellow (2025−)
Advisor	*Cooper Jacobus*	Stanford Graduate Student (2026−)
	Sixiang Yang	Stanford Graduate Student (2025−)
	Hanyue Wang	Stanford Graduate Student (2026, rotation)
	Alessandro Russo	Stanford Graduate Student (2025, rotation)
	Anita Dunsmore	Stanford Graduate Student (2025, rotation)
	Sam Goldstein	Columbia Graduate Student (2022−2025, various projects)
	John Moynihan	Columbia Undergraduate Student (2023, summer project)
	Jess Boyland	Simons-NSBP Undergraduate Scholar (2020−2021, summer project)
Coadvisor	*Luca Abu El-Haj*	Columbia Undergraduate Student (2025, summer project)
	Gemma Zhang	Harvard Graduate Student (2023−2025)
	Kristen Surrao	Columbia Graduate Student (2022−2023)
	Gerrit Farren	Cambridge Graduate Student (2020−2022)

MISCELLANEOUS

Computing Languages	Python, C++, Julia, Mathematica, Cuda
Codes Developed	PolySpec, PolyBin3D, Encore, NPCFs.jl, Class-PT, Spectra-Without-Windows, EffectiveHalos, Hipster, RascalC
Teaching	6 years of online tutoring (high-school to Masters level) Teaching assistant for Princeton introductory astronomy class (AST203) TEFL qualification in English teaching
Other	DipABRSM in Music Performance (Distinction)

REFERENCES

Prof. J C Hill
(Postdoctoral Mentor)
Columbia University
914 Pupin Hall
New York, NY 10027
Tel: +1 (212) 854-7815
jch2200@columbia.edu

Prof. D N Spergel
(PhD Advisor)
Simons Foundation
160 5th Ave.
New York, NY 10010
Tel: +1 (609) 258-3589
dspergel@simonsfoundation.org

Prof. M Zaldarriaga
(PhD Advisor)
Institute for Advanced Study
1 Einstein Drive
Princeton, NJ 08540
Tel: +1 (609) 734-8058
matiasz@ias.edu